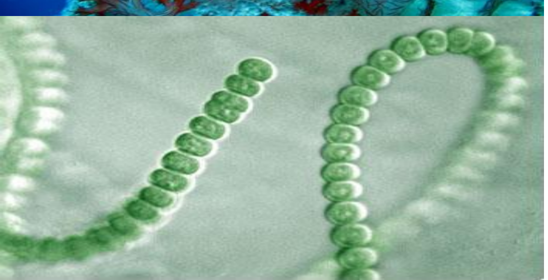




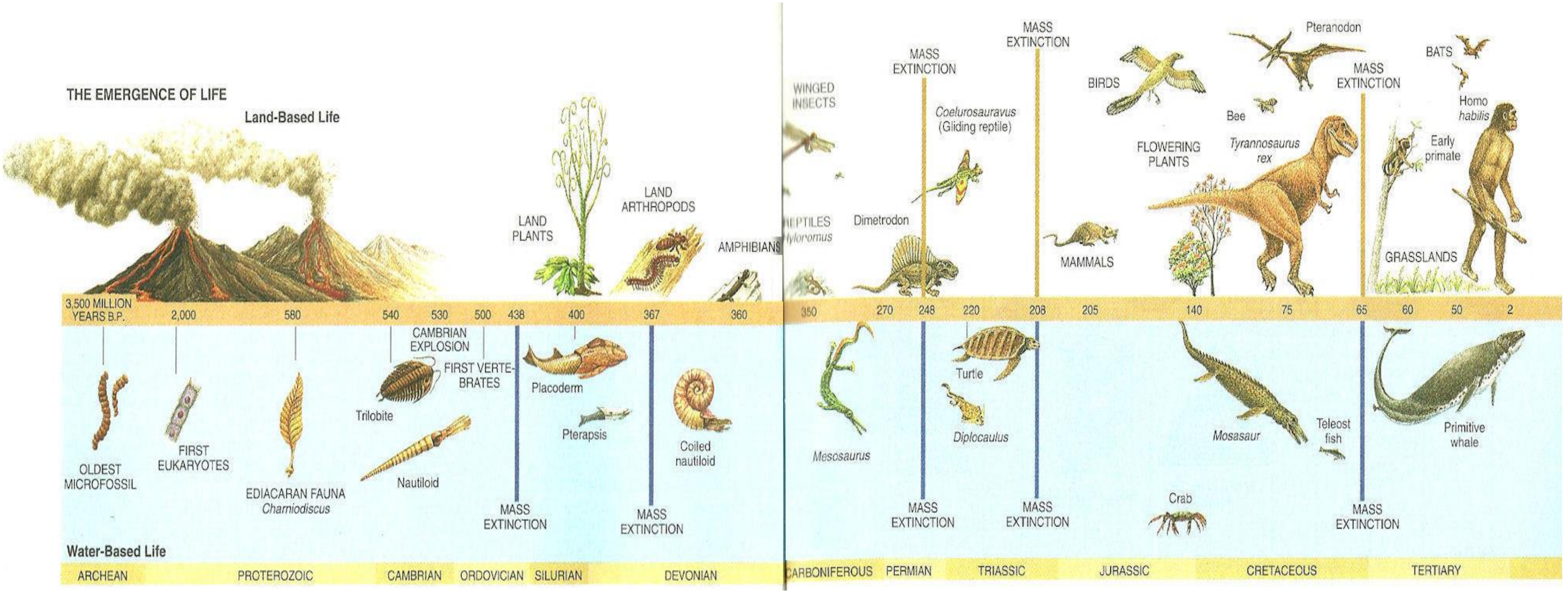


Marine Biology
海洋生物学

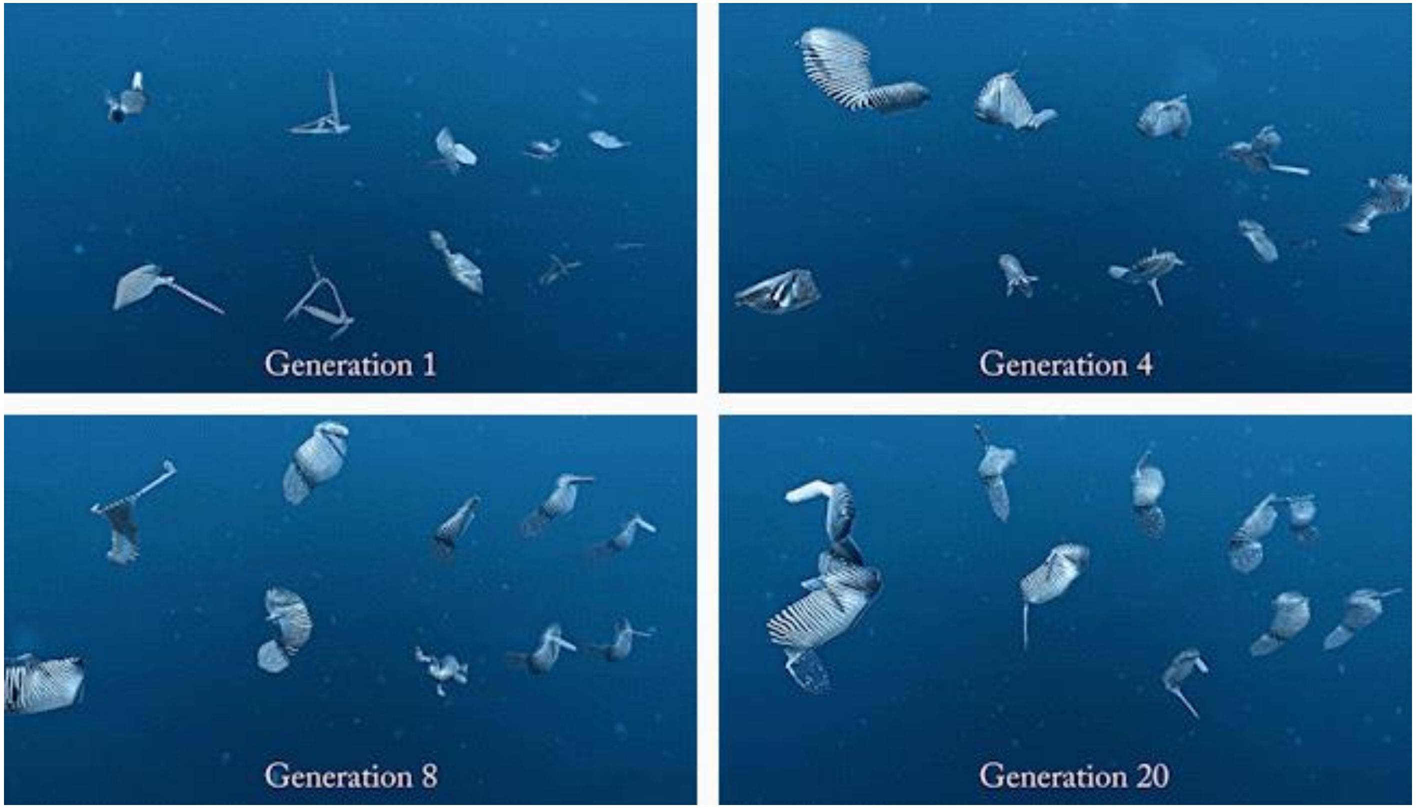
Yanan Di (邸雅楠)
diyanan@zju.edu.cn
Institute of Marine Biology and Pharmocology
Ocean College, Zhejiang University

Part 4 Evolution and Natural Selection

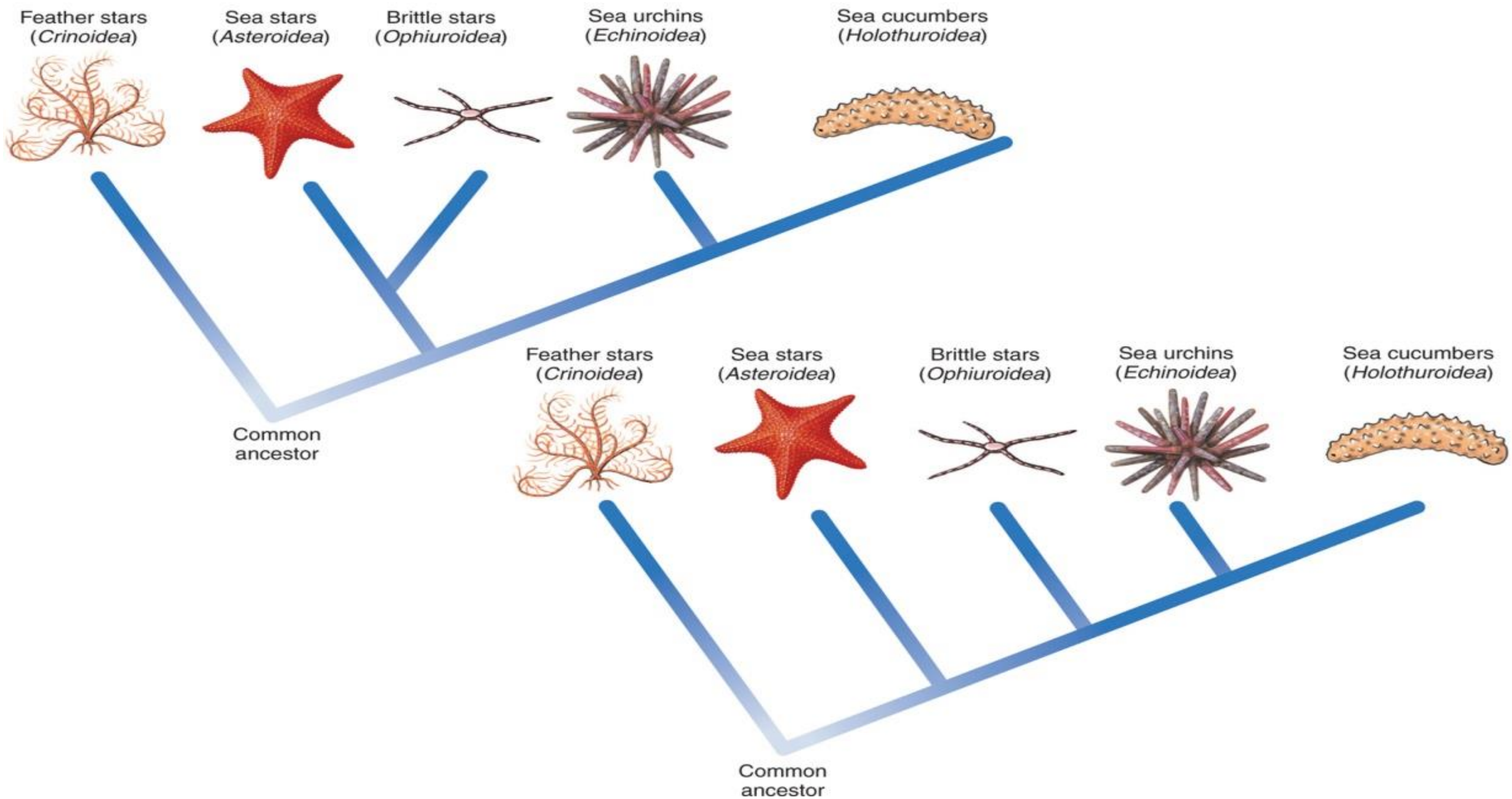
- **Evolution** is defined as a change in the genetic make-up of a population over time
- In the wild, any genetically derived traits (such as faster swimming or above-average intelligence) can give one individual survival advantage over others in his/her population. These advantages can be translated into reproductive advantage



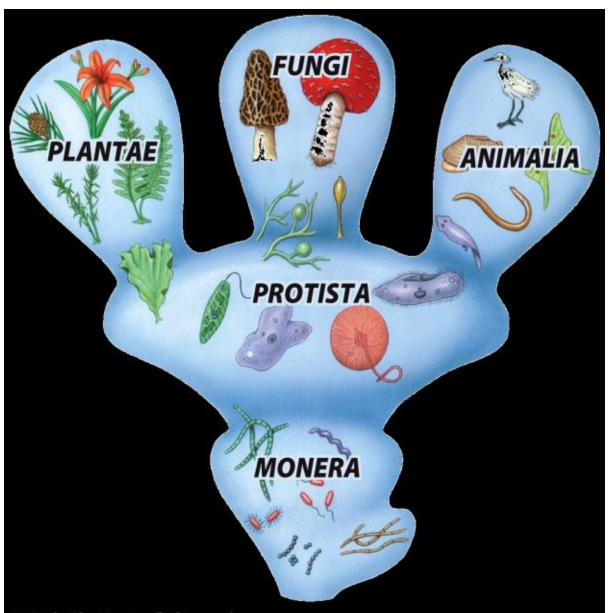
- **Natural selection** strengthen the gene pool of a species by eliminating less advantageous traits through lack (or reduction) of reproductive events in these individual
- If one organism is better survivor, more of their gametes will make it into the next generation in a population. Those individuals that are less advantaged may not survive to reproduce or will reproduce less



- **Phylogenetics** （系统发生学） is defined as the study of evolutionary relationships (relatedness) in organisms
- Biologists may use many factors to determine the relatedness of organisms such as structure, reproductive patterns, embryological or larval development, fossils, behavior or DNA/RNA



- **Taxonomy** is the science of classifying and naming organisms, based on a variety of methods including DNA and protein analysis, comparing embryos, looking at the fossil record and comparing internal and external body structures
- Taxonomy uses several levels of classification shown below from the largest (most species inclusive) to the smallest (only one species): **Domain or Kingdom → Phylum → Class → Order → Family → Genus → Species**



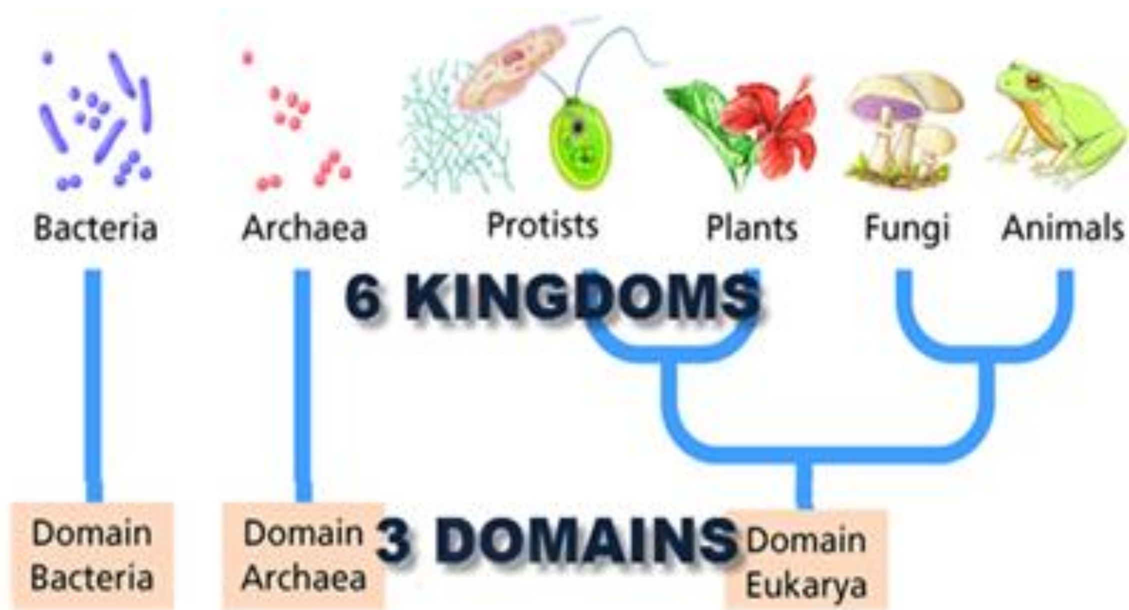


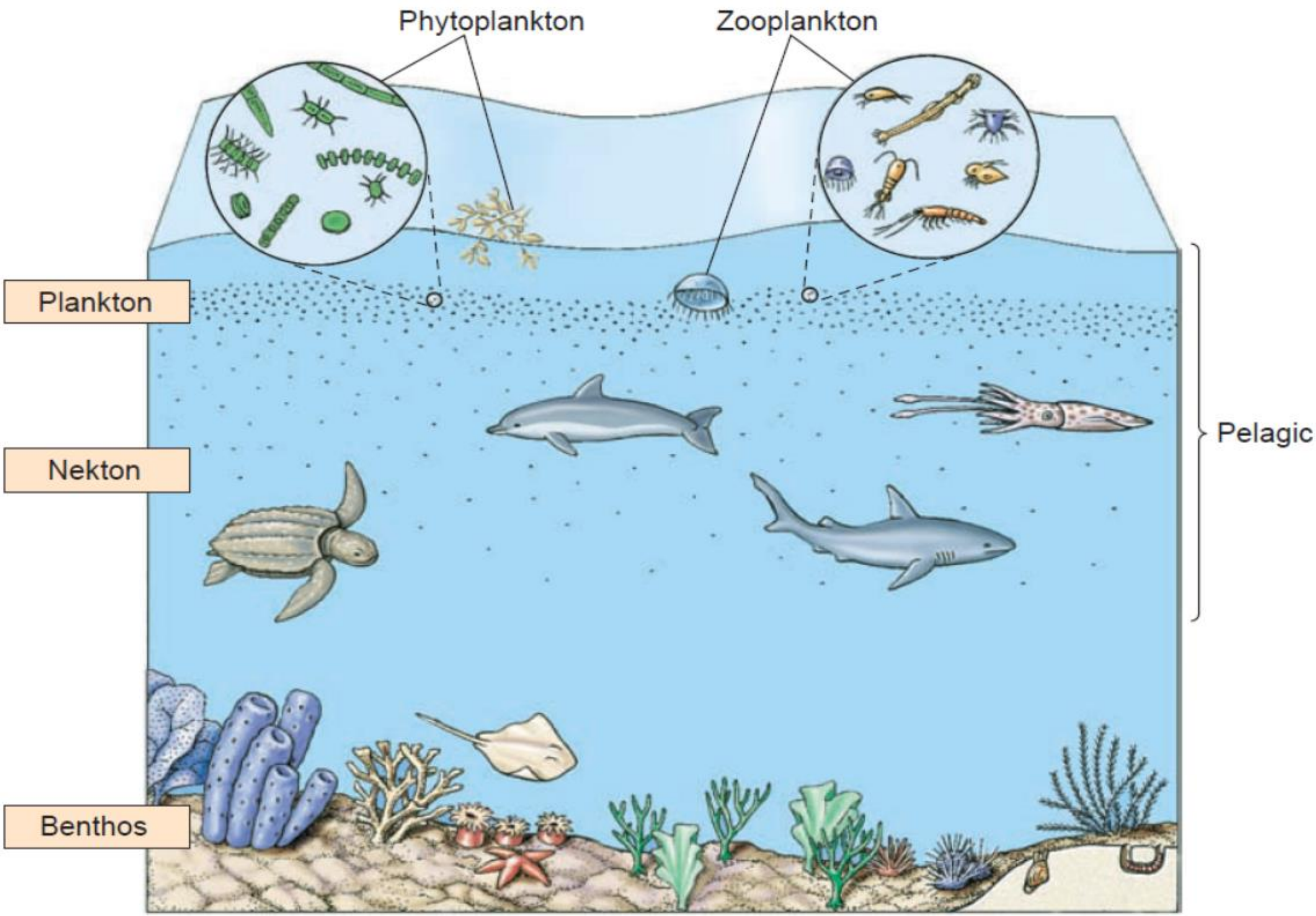
Table 4.3 Biological Classification of Humans *(Homo sapiens)* and the Bottlenose Dolphin *(Tursiops truncatus)*

	Taxonomic Level	Example
Human		
	Domain	Eukarya
	Kingdom	Animalia
	Phylum	Chordata
	Class	Mammalia
	Order	Primates
	Family	Hominidae
	Genus	*Homo*
	Species	*Homo sapiens*
Dolphin		
	Domain	Eukarya
	Kingdom	Animalia
	Phylum	Chordata
	Class	Mammalia
	Order	Cetacea
	Family	Delphinidae
	Genus	*Tursiops*
	Species	*Tursiops truncatus*

Tursiops truncatus, the bottlenose dolphin.

Note: The term "division" is used in place of "phylum" in the classification of plants.

Marine organisms:different classification



1. **<u>Nekton:</u>** Organisms that swim strongly enough to move against the current, including marine mammals, fish, cephalopods and reptile

Marine mammals—Seal（海狮）

Marine fish—Shark（鲨鱼）

Cephalopods—Octopus（章鱼）

Reptile—Sea Turtle（海龟）

2. **<u>Plankton:</u>** Organisms that drift with the water, including zooplankton and phytoplankton

- Zooplankton: the heterotrophic animal and protozoan component of plankton

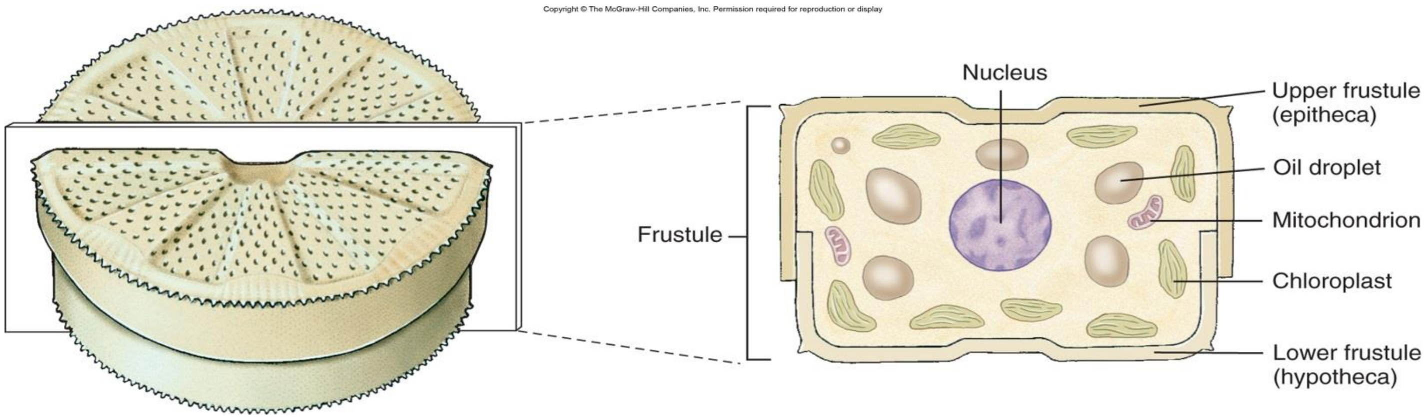
Copepod
桡足类

Jellyfish
水母

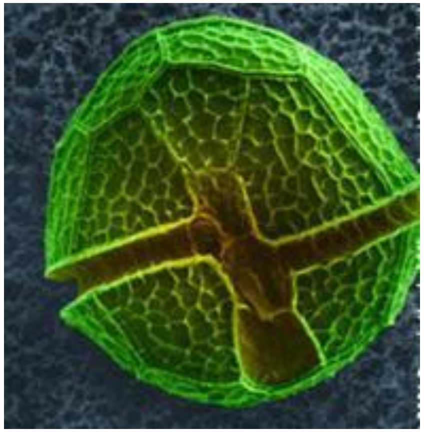
Krill
磷虾

Foraminiferans
孔虫

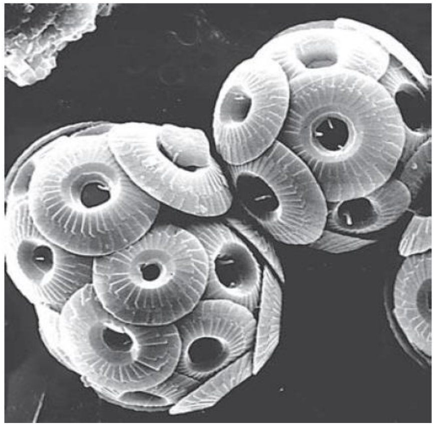
- Phytoplankton: the photosynthetic component of plankton consisting primarily of **single-celled algae** (**microalgae**) and bacteria



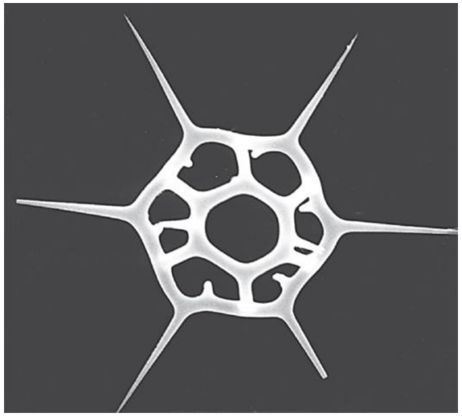
Diatom—硅藻



Dinoflagellates—甲藻



Coccolithophorids—球石藻



Sillicoflagellate—硅鞭藻

3. **Benthos:** Organisms that live on, in, or near the seabed, including phytobenthos and zoobenthos

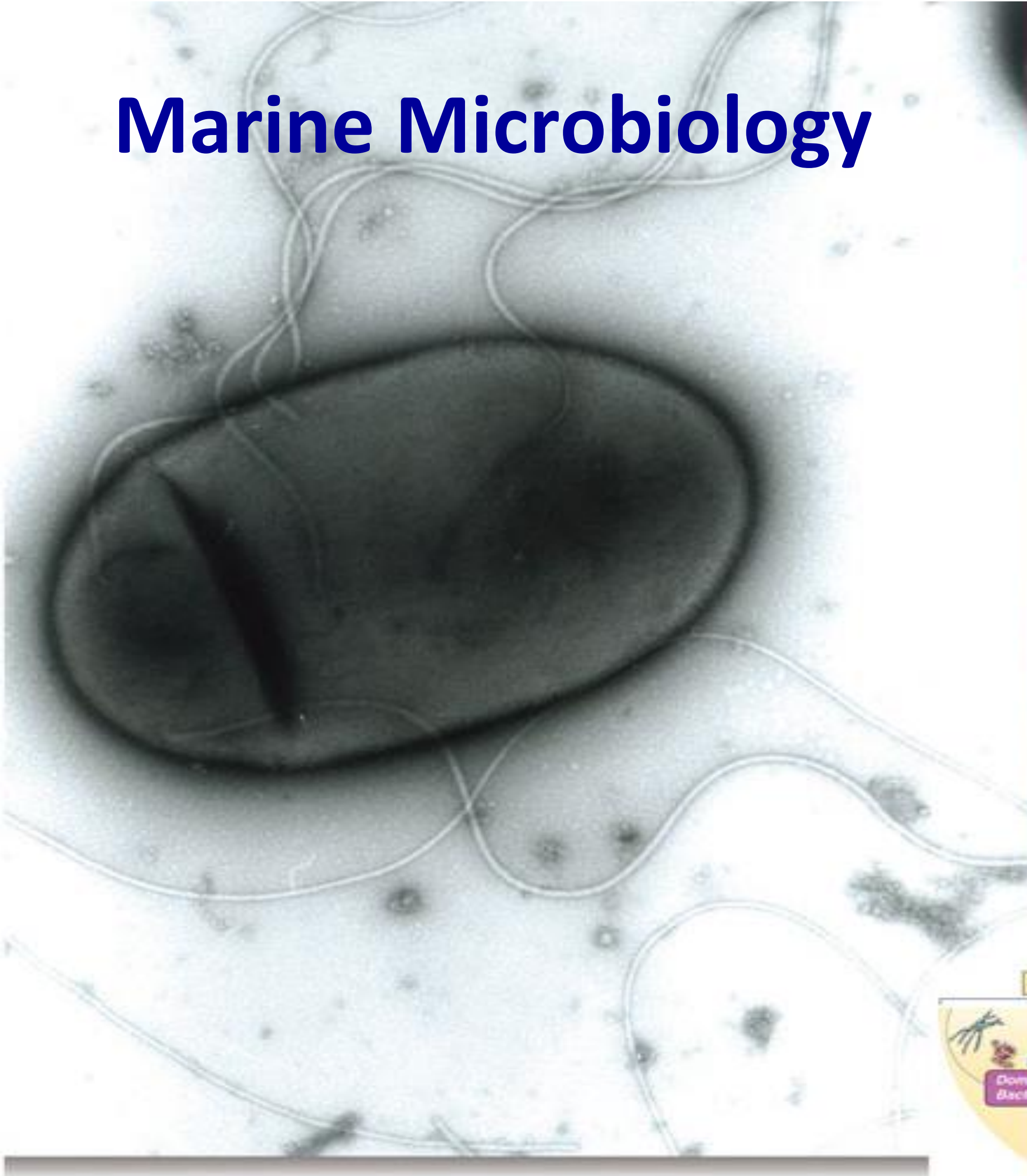
- Phytobenthos : the photosynthetic component of benthos consisting primarily of algae and bacteria
- Zoobenthos: the heterotrophic animal and protozoan component of benthos

Brow alga
褐藻（海木耳）

Red alga
红藻（紫菜）

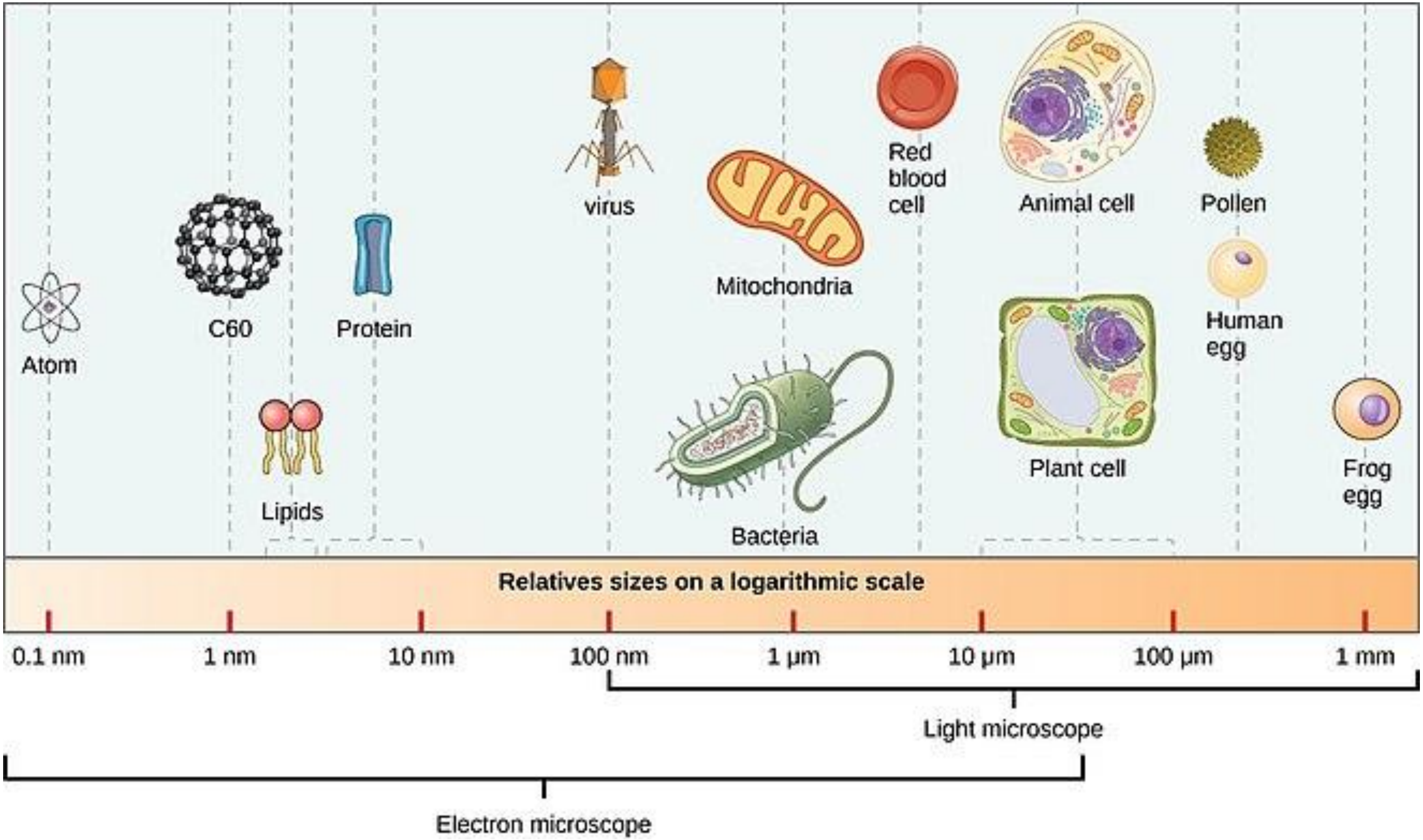
Amphipods—片脚类动物
Polychaete worm —多毛蠕虫
Snail —螺

Marine Microbiology



The Microbial World

Marine Microorganism



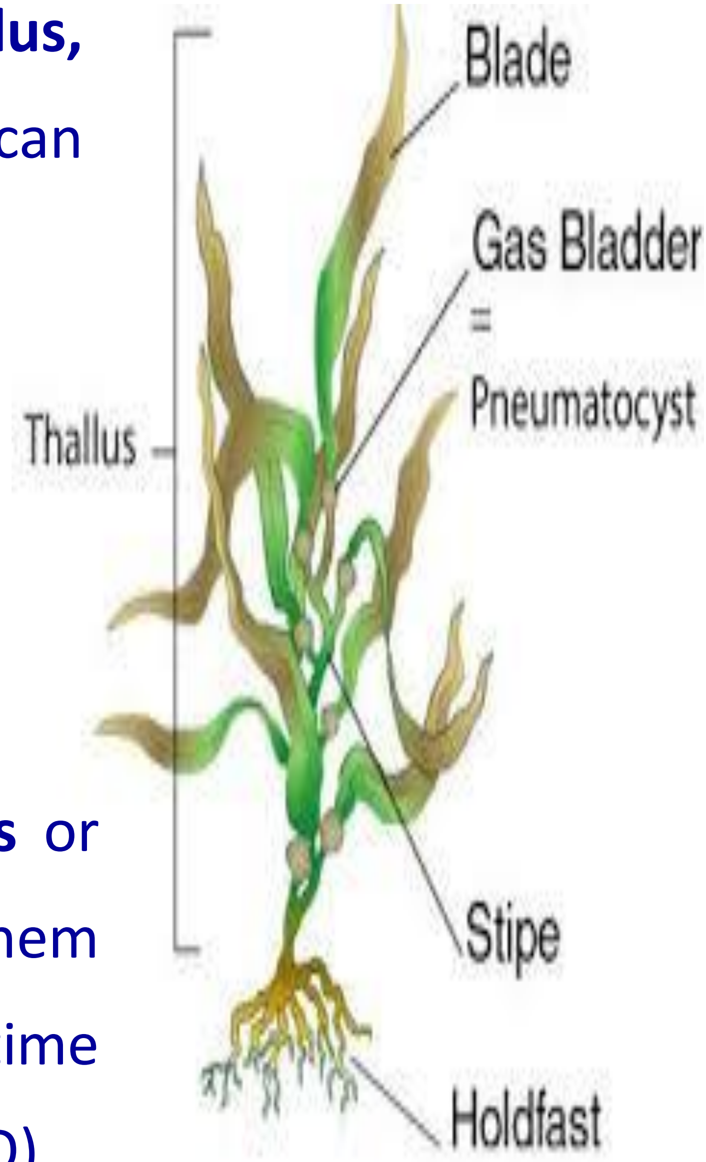
Multi-cellular Primary Producers: Seaweeds and Plants

Section 1: Marine Algae (Multi-cellular algae)

- Marine algae are important primary producers (photosynthetic), 11 phylum (**钱树本, 2005**)
- The photosynthesis rate of algae is in the range of 6-8%, which is significantly higher than the land plant (1.8-2.2%)
- These algae are called by a generic term "**seaweeds**" or more formal term **macroalgae**
- In marine ecosystem, 90-96% products are produced by microalgae, only 4-10% are produced by macroalgae
- While some are thought to be the pre-cursors of plants, algae do not have the same advanced structures seen in plants

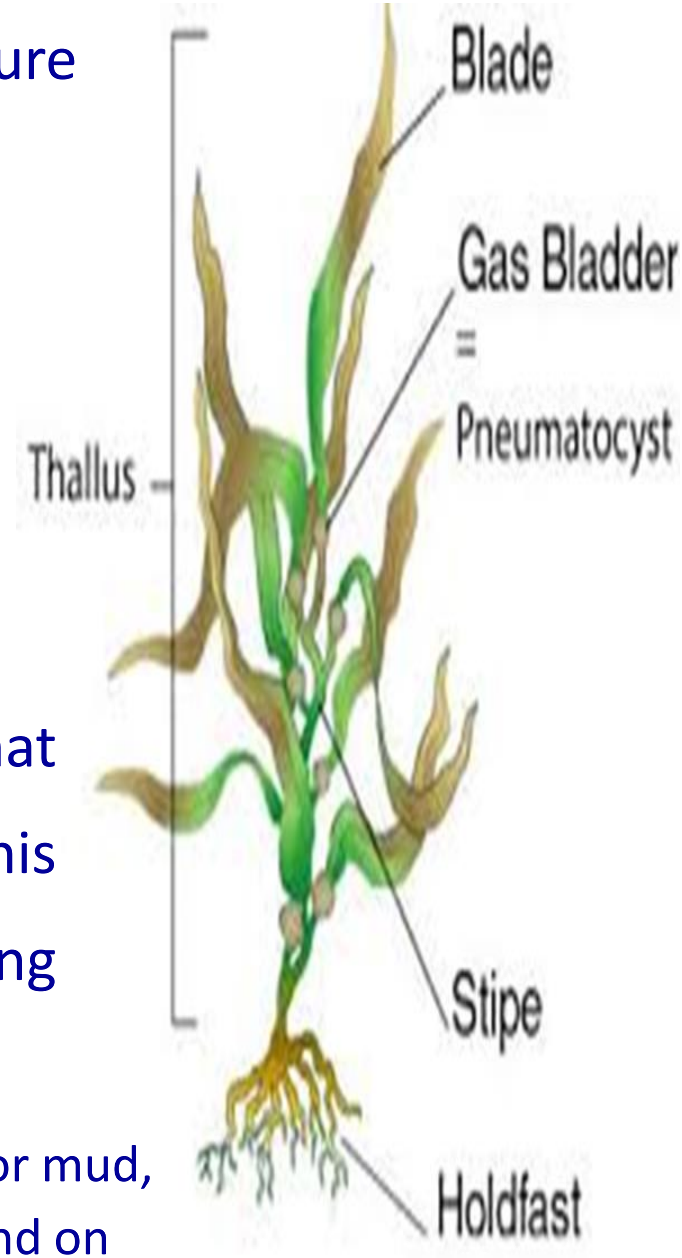
1. Basic structure of Macroalgae or Seaweeds

- The complete body is called the **thallus,** Usually, all regions of the thallus can photosynthesize
- The leaf-like flattened portions are called **blades**, the main photosynthetic region. These are not true leaves because they lack veins
- Most seaweeds have **gas-filled bladders** or **floats** or **Pneumatocyst** that will help them maximize sunlight exposure. Sometime these floats contain carbon monoxide (CO)

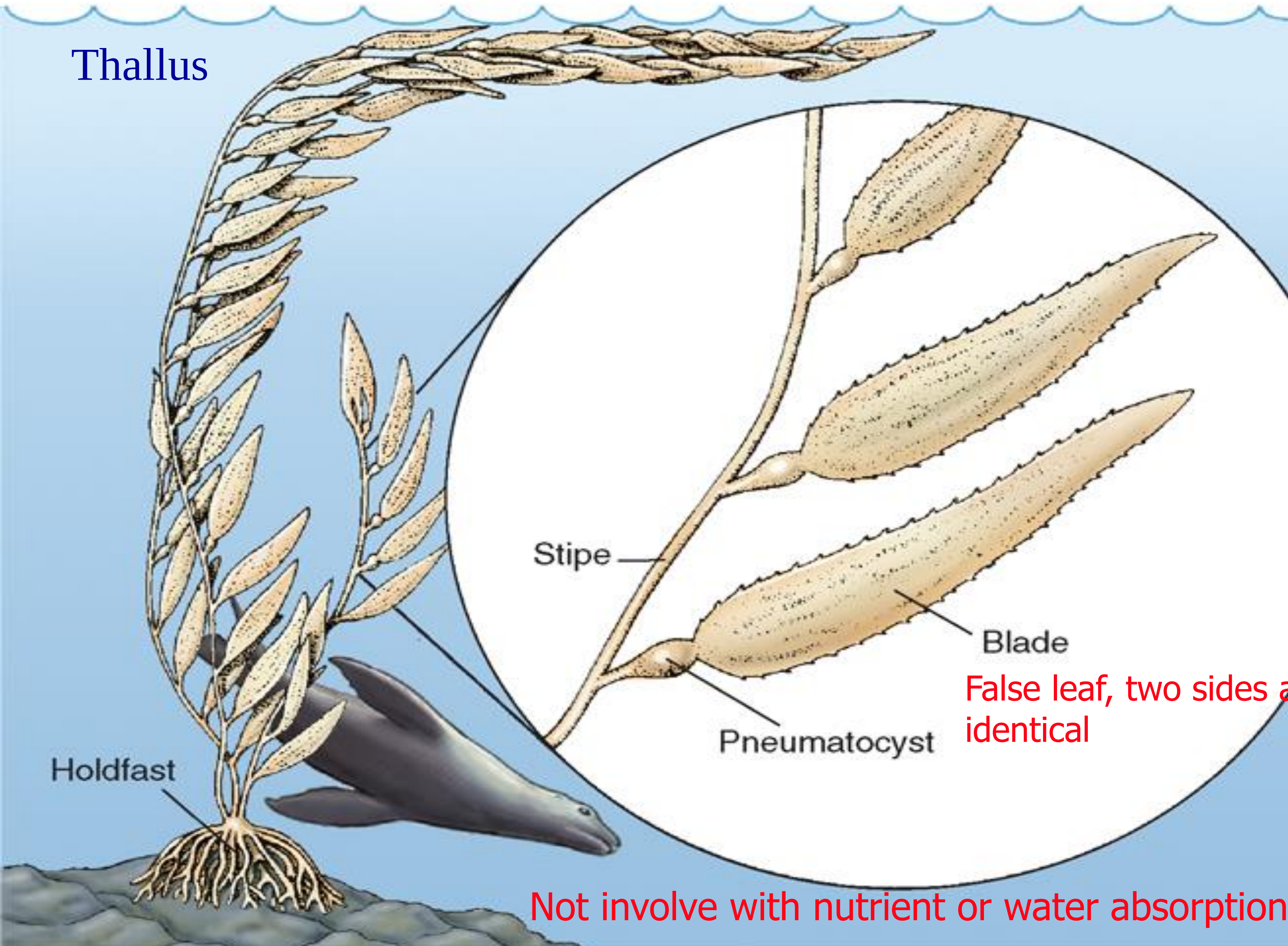


- Some seaweeds have a stem-like structure called the **stipe**, BUT not found on all seaweeds
  1. The stipe provides support and can be long and tough, as in the Giant Kelp
  2. The stipe also allows a place for the attachment of the blades
- A **holdfast** is a root-like structure that holds the seaweed to the bottom. This structure does not aid in gathering nutrients

The holdfast does not penetrate through sand or mud, so like Sea Palm, most macroalgae are only found on hard sediments



All parts of the thallus can photosynthesize as long as it has Chla



False leaf, two sides are identical

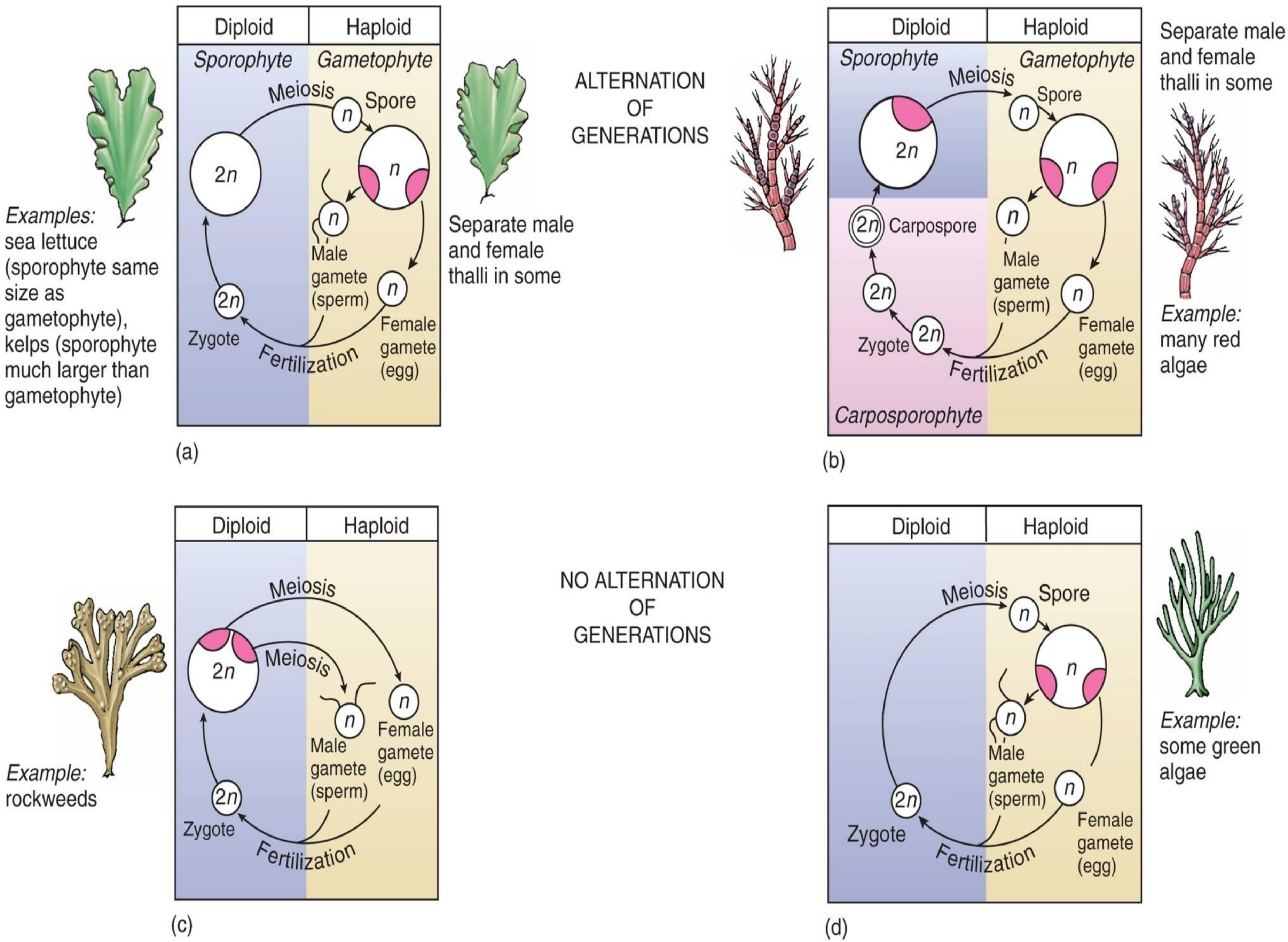
Not involve with nutrient or water absorption



2. Life History of Marine Algae

2.1 Sexual life history——algae exhibit an **alteration of generations**:

- A **gametophyte** (配子体) stage produces gametes (eggs and sperm) that will fuse to become a **zygote (合子)**
- The zygote then develops into a second stage, the **sporophyte (孢子体)**, that produces **spores (孢子)**
- The spores develop into the gametophyte stage and the cycle begins again (incidentally, plants are exhibit the same alternation of generations)



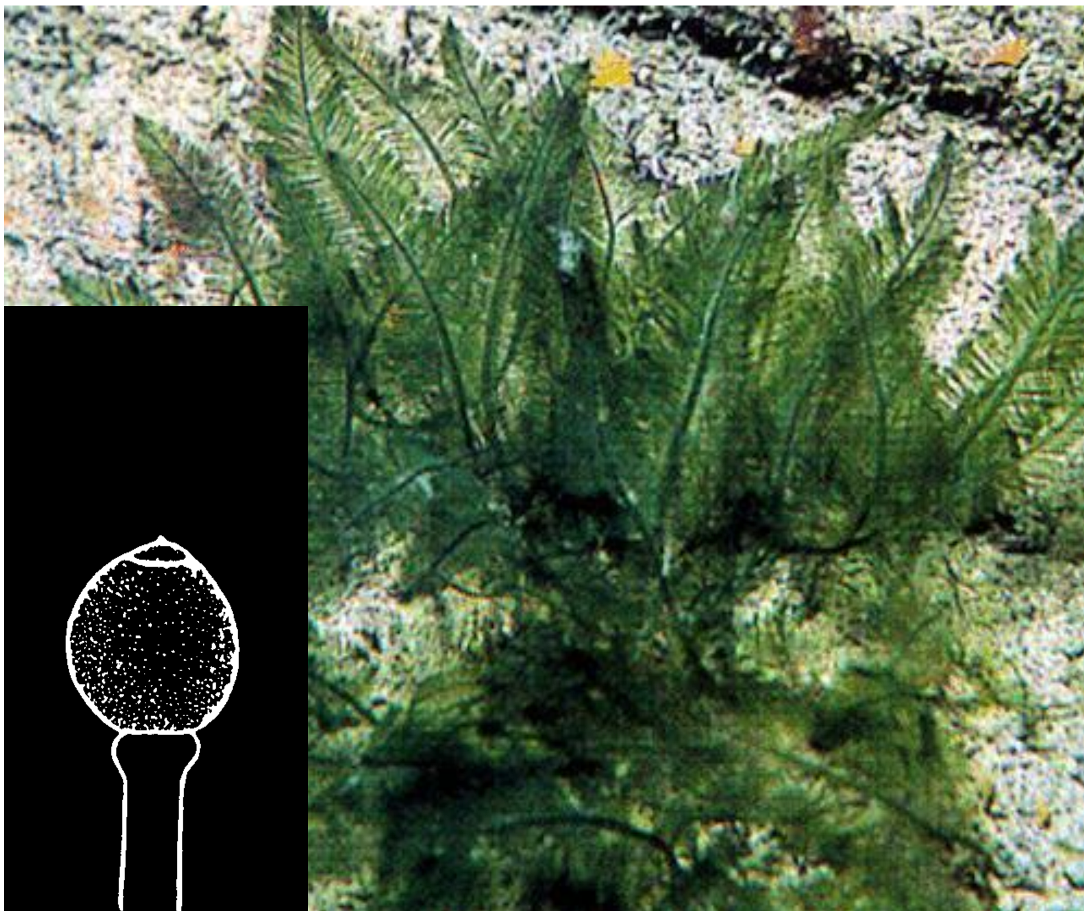
Diploid
Haploid
Sporophyte
Gametophyte
Meiosis
Spore
n
2n
n
n
Male gamete (sperm)
n
2n
Zygote
Fertilization
Female gamete (egg)
Examples:
sea lettuce (sporophyte same size as gametophyte), kelps (sporophyte much larger than gametophyte)
Separate male and female thalli in some
(a)
ALTERNATION OF GENERATIONS
Diploid
Haploid
Sporophyte
Gametophyte
Meiosis
Spore
n
2n
n
n
2n
Carpospore
Male gamete (sperm)
2n
n
Zygote
2n
Fertilization
Female gamete (egg)
Carposporophyte
Separate male and female thalli in some
Example:
many red algae
(b)
Diploid
Haploid
Meiosis
2n
Meiosis
n
n
Male gamete (sperm)
Female gamete (egg)
2n
Zygote
Fertilization
Example:
rockweeds
(c)
NO ALTERNATION OF GENERATIONS
Diploid
Haploid
Meiosis
Spore
n
n
n
Male gamete (sperm)
n
2n
Zygote
Fertilization
Female gamete (egg)
Example:
some green algae
(d)

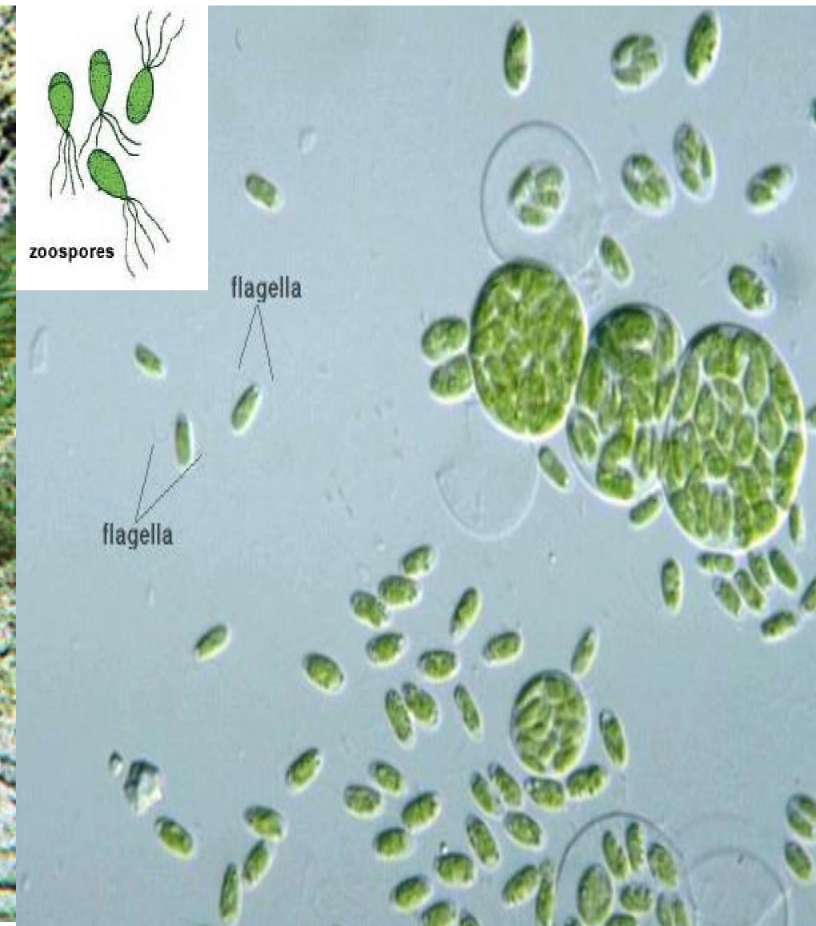
2.2 Asexual life history:

- Sexual reproduction is expensive both energetically and physiologically
- Many algae also reproduce asexually by a means called vegetative growth, e.g. fragments, spores, or buds
- Asexual reproduction make sure an algae reproduces new individuals that are genetically identical to the parent algae
- Sometimes algae use both sexual and asexual reproduction depending on environmental conditions

<u>Spores:</u> cells are specialized for dispersing to new location or survive through unfavorable condition

<u>Zoospore:</u> spores have flagella for movements





Homework:

Watch the videos provided, questions will be asked on next lecture